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SUPERSEDING
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MILITARY SPECIFICATION

SWITCHES, ROTARY: 28 VOLT DC

This specification is approved for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers 28 volt (V) direct current (dc) rotary switches intended primarily for use in tactical vehicles (see 6.1).

1.2 Classification. Switches shall be of the following types and classes as specified (see 6.2):

Type I	- Single-pole, single throw switches.
Type II	- Single-pole, multi-throw switches.
Type III	- Multipole, single throw switches.
Type IV	- Multipole, multi-throw switches.
Class 1	- Waterproof.
Class 2	- Non-Waterproof.
Class 3	- Non-Waterproof, with sealed (waterproof) shaft.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: AMSTA-GDS, Warren, MI 48397-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS
FEDERAL

QQ-C-320	- Chromium Plating (Electrodeposited).
QQ-P-416	- Plating, Cadmium (Electrodeposited).
QQ-S-365	- Silver Plating, Electrodeposited; General Requirements for.
QQ-S-571	- Solder: Tin Alloy, Tin Lead Alloy and Lead Alloy.

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MIL-M-14	- Molding Plastics and Molded Plastic Parts, Thermosetting.
MIL-C-13486	- Cable, Special Purpose, Electrical, Low Tension.
MIL-F-13927	- Fungus Resistance Test; Automotive Components.
MIL-E-52798	- Enamel, Alkyd, Camouflage.
MIL-E-52835	- Enamel, Modified Alkyd, Camouflage, Lusterless.

STANDARDS
MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-193	- Painting Procedures and Marking for Vehicles, Construction Equipment and Material Handling Equipment.
MIL-STD-202	- Test Methods for Electronic and Electrical Component Parts.
MIL-STD-1184	- Electrical Components for Automotive Vehicles; Waterproofness Tests.
MIL-STD-45662	- Calibration Systems Requirements.
MS27148	- Pin Contact, Electrical Connector - No. 12, 14 and 15 AWG.
MS39060	- Switch, Rotary, 28 Volt DC, Waterproof.

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2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

NATIONAL BUREAU OF STANDARDS

Handbook H28 - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D.C. 20025.)

(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. Unless otherwise specified (see 6.2), the contractor shall furnish switches which shall be subjected to first article inspection (see 4.4). First article inspection samples, properly marked with identifying information shall be representative of the units to be furnished to the Government. All subsequent switches delivered to the Government shall conform to these samples in all of their pertinent physical and performance attributes.

3.2 Design and construction. Switches and assemblies shall conform to this specification and to MS39060, MS27148, or, when specified, other applicable standards or drawings (see 4.6.1 and 6.2).

3.2.1 Materials. Materials shall be as specified herein, in applicable specifications and drawings or standards. Materials not specifically designated shall be appropriate for use in the assembly conforming to the requirements herein and in applicable specifications (see 4.6.1).

3.2.1.1 Dissimilar metals. Except where necessary to complete an electrical circuit, contact between dissimilar metals, which would encourage galvanic action shall be avoided. Separation of dissimilar metals shall be accomplished by providing insulation between mating surfaces (see 4.6.1).

3.2.2 Clearance. Clearance between any current-carrying and other conducting parts of different potential shall be no less than 1/16 inch through air or insulation (see 4.6.1).

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3.2.3 Standard parts. Military standard parts shall be incorporated wherever applicable. Commercial standard parts may be used provided they are interchangeable with Military Standard (MS) parts without modification, and the equivalent MS part is identified by the manufacturer (see 4.6.1).

3.2.4 Threaded parts. All screw threads shall conform to National Bureau of Standards Handbook H28 (see 4.6.1).

3.2.5 Springs. Springs shall not be used to transmit current (see 4.6.1).

3.2.6 Locking devices. Lockwashers, self-locking nuts, or other approved locking devices shall be incorporated where specified or where required to prevent loosening of components (see 4.6.1).

3.2.7 Solder. Solder shall conform to composition Sn50, Sn60, or Sb5 of QQ-S-571. A rosin flux shall be used (see 4.6.1).

3.2.8 Plastic. Molded insulation material shall conform to type MFE requirements of MIL-M-14 (see 4.6.1).

3.2.9 Cable. Cable shall conform to MIL-C-13486 (see 4.6.1).

3.2.10 Interchangeability. The switch shall be interchangeable to the extent that external dimensions shall be within the envelope outline of the applicable drawing or standard. Electrical connectors, shaft and mounting dimensions and location shall conform to the detail requirements of the applicable drawing (see 4.6.2).

3.2.11 Weight. The weight of the switch shall be in accordance with the weight requirements of the applicable drawing (see 4.6.2).

3.2.12 Terminals. Tip dimensions of round terminals for snap-on connectors shall conform to MS27148-3 (see 3.2 and 4.6.2).

3.3 Performance. The switch shall perform as specified herein when energized at a rated load of 28 V dc.

3.3.1 Voltage drop. When connected as in use and energized, the voltage drop between terminals shall be no greater than 150 millivolts (see 4.6.1 and 6.4.1).

3.3.2 Overload. The switch shall perform as specified in 3.3.1 after cycling while connected to loads 40 percent greater than rated loads (see 4.7.2).

3.3.3 Endurance. After 50,000 cycles of operation at rated loads, the voltage drop shall be no greater than 500 millivolts. The torque required to operate the switch shall be no less than 30 ounce-inches (see 3.3.5 and 4.7.3).

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3.3.4 Dielectric strength. The switch shall perform as specified in 3.3.1 and shall evidence no burning, charring, loosening, arcing or other damage to the insulation after operating in each "on" and "off" position with 1000 V root mean square (rms) applied (see 4.7.4).

3.3.5 Torque. The switch shall operate with a torque of not less than 30 nor more than 70 ounce-inches. The switch shall perform as specified herein and evidence no damage after operation with a 3 foot-pound torque (see 4.7.5).

3.3.6 Terminal strength. Terminals of the switch shall perform as specified in 3.3.1 and 3.3.4 after withstanding perpendicular forces and a longitudinal pull of 25 pounds each (see 4.7.6).

3.3.7 Shock resistance. The switch shall perform as specified in 3.3.1 after exposure to shock. The switch shall evidence no mechanical or electrical malfunction (opening or closing of circuit) during and after exposure to 3 sawtooth shock pulses, each of 50 gravities (g) peak amplitude for 7 milliseconds (ms) duration, in each of three orthogonal axes for a total of 18 shock pulses (see 4.7.7).

3.3.8 Vibration resistance. The switch shall perform as specified in 3.3.1 and 3.3.5 after exposure to vibration for one hour in each of three orthogonal axes for a total of three hours. The vibration shall be an harmonic motion having an amplitude of 0.03 inch (0.06-inch total excursion) in a frequency sweep from 10 to 55 and return to 10 Hertz (Hz) in 1 minute (see 4.7.8).

3.4 Environmental conditions. The switch shall meet the performance specified in 3.3.1 after exposure to the range of environmental conditions expected for military usage (see 4.8).

3.4.1 Salt spray resistance. The switch shall perform per 3.3.1 during and after exposure to a fog of atomized salt water of 5 percent salt solution for 200 hours (see 4.8.1).

3.4.2 Sand and dust resistance. The switch shall perform as specified in 3.3.1 after exposure to sand and dust in an air velocity of 2500 $\pm$ 500 feet per minute (see 4.7.2).

3.4.3 Fungus resistance. The switch shall perform as specified in 3.3.1 after exposure to fungus growth for 90 days (see 4.8.3).

3.4.4 Waterproofness. The switch shall be mechanically and electrically operable during and after submersion in saline solution. During submersion, switches shall not leak (see 4.8.4).

3.4.5 Temperature resistance. The switch shall perform as specified in 3.3.1 and 3.3.3 during and after exposure to extreme temperatures (see 4.8.5).

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3.4.5.1 Low temperature resistance. The switch shall perform as specified herein after exposure at minus 65 ± 5 degrees Fahrenheit (°F). During exposure to extreme cold the switch shall perform as specified in 3.3.1 and the operational torque shall not exceed by more than 40 percent the operational torque of 3.3.5 (see 4.8.5.1).

3.4.5.2 High temperature resistance. The switch shall perform as specified in 3.4.5 during and after exposure at plus 165 ± 5°F (see 4.8.5.2).

3.5 Finish.

3.5.1 Color. Unless otherwise specified (see 6.2), switches shall be Forest Green conforming to MIL-E-52798 or MIL-E-52835 (see 4.6.2).

3.5.2 Painting. Cleaning, conditioning, and painting shall conform to the applicable requirements of MIL-STD-193 (see 4.6.1).

3.5.3 Chemical and electro-chemical treatment. Protective or foundation chemical and electro-chemical treatments of metals and alloys, shall conform to the applicable requirements of MIL-STD-193 (see 4.6.1).

3.5.4 Plating. When specified (see 6.2) exposed unprotected metal surfaces shall be cadmium plated in accordance with class 2, type II of QQ-P-416. Silver plating shall conform to QQ-S-385. Chromium plating shall conform to class 2 of QQ-C-320 with a film thickness of no less than 0.0002 inch (see 4.6.1).

3.6 Marking.

3.6.1 Identification marking. Identification marking shall be in accordance with MIL-STD-130 and as a minimum shall include the following (see 4.6.2):

Military Part Number
Manufacturer's Identification including FSCM
Manufacturer;s Serial Number or date of
Manufacture (month, year, e.g. Feb 67).

3.6.2 Index marking. Index marking shall be permanent and shall conform to MIL-STD-130 and the applicable military standard or drawing (see 4.6.2).

3.7 Workmanship. Workmanship shall be such as to assure a product free of rust, burrs, scratches, chips and sharp edges (see 4.6.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order (see 6.2), the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the

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inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform or witness any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply wih all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Inspection equipment. Unless otherwise specified in the contract (see 6.2), the contractor is responsible for the provision and maintenance of all inspection equipment necessary to assure that supplies and services conform to contract requirements. Inspection equipment must be capable of repetitive measurements to an accuracy of 10 percent of the measurement tolerance. Calibration of inspection equipment shall be in accordance with MIL-STD-45662 and as shown in table I.

TABLE I. Measuring instrument tolerance.

Measuring instrument	Accuracy
Voltmeter DC	± 0.25 percent
Ammeter DC-	± 0.50 percent
Thermometer	± 1.0 percent

4.2 Classification of inspection:

a. First article inspection (see 4.4).
b. Quality conformance inspections (see 4.5).
    1. Examination (see 4.5.2).
    2. Tests (see 4.5.3).
c. Control tests (see 4.5.4).

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4.3 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be conducted under the following conditions:

a. Air temperature 77 ± 15°F
b. Barometric pressure 725 + 50 millimeter (mm) mercury (Hg)
- 75
c. Relative humidity 50 ± 30 percent

4.4 First article inspection. Unless otherwise specified (see 6.2), the Government shall select six switches produced under the production contract for first article inspection. First article testing shall be conducted by the contractor under Government surveillance, and shall consist of examination for classification of defects specified in 4.5.2 and testing specified in table II in the order listed on table III. Approval of the first article sample by the Government shall not relieve the contractor of his obligation to supply regulator assemblies that are fully representative of those inspected as a first article sample. Any changes or deviation of the production units from the first article sample shall be subject to the approval of the contracting officer.

4.4.1 First article inspection failure. Test item deficiencies during, or as a result of, the first article test, shall be cause for rejection of the items until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiency. Any deficiency found during, or as a result of the first article test, shall be evidence that all items already produced prior to completion of the first article test are similarly deficient unless contrary evidence satisfactory to the contracting officer is furnished by the contractor. Such deficiencies on all items shall be corrected by the contractor. The Government shall not accept products until first article testing is completed to the satisfaction of the Government.

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TABLE II. Classification of inspections.

Title	Requirement	Inspection	Qualification	Quality conformance		Control
				Examination	Tests	
Materials and construction	3.2 thru 3.2.9 and 3.5.2 thru 3.5.4	4.6.1	X			
Defects	3.2.10 thru 3.2.12, 3.5.1, and 3.6.1 thru 3.7	4.6.2	X	X		
Voltage drop	3.3.1	4.7.1	X			X
Overload	3.3.2	4.7.2	X			X
Endurance	3.3.3	4.7.3	X			X
Dielectric strength	3.3.4	4.7.4	X			
Torque	3.3.5	4.7.5	X		X	
Terminal strength	3.3.6	4.7.6	X		X	
Shock resistance	3.3.7	4.7.7	X			
Vibration resistance	3.3.8	4.7.8	X			X
Environmental conditions	3.4	4.8	X			
Salt spray resistance	3.4.1	4.8.1	X			
Sand and dust resistance	3.4.2	4.8.2	X			
Fungus resistance	3.4.3	4.8.3	X			
Waterproofness	3.4.4	4.8.4	X			X
Temperature resistance	3.4.5	4.8.5	X			
Low temperature resistance	3.4.5.1	4.8.5.1	X			
High temperature resistance	3.4.5.2	4.8.5.2	X			

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TABLE III. Order of first article tests.

Sample number	Paragraph number	Test sequence
1 and 2	4.7.1	Voltage drop
	4.7.6	Terminal strength
	4.7.7	Shock resistance
	4.7.8	Vibration resistance
	4.8.4	Waterproofness
	4.7.1	Voltage drop
	4.8.3	Fungus resistance
	4.7.4	Dielectric strength
	4.7.5	Torque
	4.7.1	Voltage drop
3 and 4	4.7.1	Voltage drop
	4.7.5	Torque
	4.7.1	Voltage drop
	4.8.5, 4.8.5.1 and 4.8.5.2	Temperature resistance
	4.7.2	Overload
	4.8.1	Salt spray resistance
	4.7.4	Dielectric strength
5 and 6	4.7.1	Voltage drop
	4.7.4	Dielectric strength
	4.8.2	Sand and dust resistance
	4.7.3	Endurance
	4.7.4	Dielectric strength

4.5 Quality conformance inspection.

4.5.1 Sampling.

4.5.1.1 Lot formation. An inspection lot shall consist of all the switches of one type and part number, from an identifiable production period, from one manufacturer, submitted at one time for acceptance.

4.5.1.2 Sampling for examination. Samples for quality conformance examination shall be selected in accordance with general inspection level II of MIL-STD-105.

4.5.1.3 Sampling for testing. A representative sample shall be selected in accordance with level S-3 of MIL-STD-105.

4.5.2 Examination.

4.5.2.1 Acceptable quality level. Each sample selected in accordance with 4.5.1.2 shall be examined to determine conformance to the following acceptable quality levels (AQL).

Classification	AQL
Major	1.0
Minor	2.5

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4.5.2.2 Classification of defects. For examination purposes, defects shall be classified as specified in table IV.

TABLE IV. Classification of defects.

Category	Defect	Method of examination
Critical	None	
Major	AQL 1.0% Defective	
101	Incorrect dimensions affecting installation (see 3.2.10).	SIE 1/
102	Incorrect dimensions of terminal tips (see 3.2.12).	SIE
Minor	AQL 2.5% Defective	
201	Incorrect dimensions not affecting installation (see 3.2.10).	SIE
202	Over or under weight (see 3.2.11).	SIE
203	Faulty finish or wrong color (see 3.5 and 6.2).	Visual
204	Improper or incomplete marking (see 3.6.1 and 3.6.2).	Visual
205	Faulty workmanship (see 3.7).	Visual

1/ SIE = Standard Inspection Equipment.

4.5.3 Quality conformance test. Samples selected in accordance with 4.5.1.3 shall be subjected to the tests specified in table II using an AQL of 6.5 on the basis of percent defective. These samples shall be tested in the following sequence: 4.7.5 and 4.7.6.

4.5.4 Control tests. Control tests shall be conducted on two switches from each lot of 500 switches consecutively produced, except that not more than four nor less than two switches shall be subjected to control testing in any 90 day period. The control test samples shall be identified as to production period and, after having been examined for the defects specified in 4.5.2.2, they shall be subjected to the control tests specified in table II in the following order: 4.7.1 and again after each test, 4.7.8, 4.8.4, and 4.7.3.

4.5.5 Failure. Failure of any switches to pass any of the specified quality conformance or control tests shall be cause for the Government to refuse acceptance of the production quantity represented, until action taken by the contractor to correct defects and prevent recurrence has been approved by the Government.

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4.6 Methods of inspection.

4.6.1 Materials and construction. Conformance to 3.2 through 3.2.9, and 3.5.2 through 3.5.4 shall be determined by inspection of contractor records providing proof or certification that design, construction, processing, and materials conform to requirements. Applicable records shall include drawings, specifications, design data, receiving inspection records, processing and quality control standards, vendor catalogs and certifications, industry standards, test reports, and rating data.

4.6.2 Defects. Conformance to 3.2.10 through 3.2.12, 3.5.1, and 3.6.1 through 3.7 shall be determined by examination for the defects listed in table IV. Examination shall be visual, tactile, or by measurement with standard inspection equipment.

4.7 Inspection for performance.

4.7.1 Voltage drop. To determine conformance to 3.3.1, the switch shall be energized by a 28 V dc source and connected to the rated loads specified on the applicable drawing (see 3.2). Measure the terminal-to-terminal voltage. Verify that the voltage drop is not greater than 150 millivolts.

4.7.2 Overload. To determine conformance to 3.3.2, the switch shall be energized by a 28 V dc source and connected to loads 40 percent greater than rated (lamp, resistive and inductive) loads. The switch shall be operated for 100 cycles at each specified load. For lamp loads, the load cycle shall be 1 $\pm$ 0.1 second "on" and 5 $\pm$ 0.1 seconds "off" (10 cycles per minute). The inductive load shall be 3 $\pm$ 0.1 seconds "on" and 3 $\pm$ 0.1 seconds "off" (10 cycles per minute). After cycling is completed test the switch as specified in 4.7.1.

4.7.3 Endurance. To determine conformance to 3.3.3, the switch shall be energized by a 28 V dc source and connected to the rated loads specified on the applicable drawing (see 3.2) and operated for 50,000 cycles. During testing, the switch shall show no evidence of malfunction. After the 50,000-cycle operation, measure the terminal-to-terminal voltage. Verify that voltage drop is not greater than 500 millivolts. Mount the switch as in 4.7.5. Verify that the torque required to operate the switch after the 50,000-cycle operation is not less than 30 ounce-inches.

4.7.4 Dielectric strength. To determine conformance to 3.3.4, the switch shall be tested as specified in MIL-STD-202, Method 301. Applied voltage shall be 1000 V rms at a frequency of 60 cycles per second with the switch lever in each of its "on" and "off" positions. The voltage shall be increased 400 V per second between terminals and between insulated terminals and ground for one minute on each application. Verify that there is no evidence of burning, charring, lossening, arcing or other damage to the insulation.

4.7.5 Torque. To determine conformance to 3.3.5, the switch shall be secured to a fixture simulating mounting as in intended use and fitted with an operating lever, then tested as follows:

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a. A force shall be applied to produce the torque required to move the switch from "on" or "off" position as in intended use. The operating torque shall be the average of five measurements of the torque applied for each direction of movement from each position. Verify that the torque is not less than 30 nor more than 70 ounce-inches.

b. With the lever turned to each extreme position a force shall be applied in the direction affording maximum leverage such that a torque load of 3 foot-pounds is applied to the switch. This load shall be applied gradually five times for each lever position and held for at least 10 seconds. Verify that the switch evidences no breakage, distortion or loosening of parts. After testing for both operating torque and maximum torque, verify that the voltage drop requirement is maintained by testing as specified in 4.7.1.

4.7.6 Terminal strength. To determine conformance to 3.3.6, the switch shall be mounted as in intended use and a force of 25 pounds shall be applied at the base of all terminals in four directions perpendicular to the longitude centerline of the terminals at 90° intervals. In addition a 25 pound pull shall be applied along the terminal axes centerline. Force and pull shall be applied for 10 seconds each. Verify that there is no evidence of damage, distortion or loosening of parts. Subsequently the switch shall be tested per 4.7.1 and 4.7.4 to verify that voltage drop and dielectric strength requirements are maintained.

4.7.7 Shock resistance. To determine conformance to 3.3.7, the switch shall be mounted and electrically connected in a fixture to simulate mounting in intended use. Subject the switch to three sawtooth shock pulses of 50 g peak amplitude of 7 ms duration in each of three major axes for a total of 18 shock pulses. One axis shall be through or parallel to the plane of contact movement and the lever when in "off" position. This test shall conform to MIL-STD-202, method 213. The shock-testing apparatus shall be constructed with a non-yielding base and a stationary anvil. Verify that there is no evidence of electrical or mechanical malfunction during and after this shock test.

4.7.8 Vibration resistance. To determine conformance to 3.3.8, the switch shall be mounted and electrically connected as in intended use and subjected to the vibration test specified in method 201 of MIL-STD-202, except that vibration shall be for one hour along each of three mutually perpendicular axes, one of which shall be through or parallel to the switch shaft. Verify that there is no evidence of electrical or mechanical malfunction. Subsequently the switch shall be tested for operating torque per 4.7.5(a) and for voltage drop per 4.7.1.

4.8 Environmental conditions. The switch shall be tested per 4.7.1 after exposure to the environmental tests listed herein.

4.8.1 Salt spray resistance. To determine conformance to 3.4.1, the switch shall be tested as specified in MIL-STD-202, method 101, except that the test duration shall be 200 hours. A 5 percent salt solution shall be used.

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4.8.2 Sand and dust resistance. To determine conformance to 3.4.2, the switch shall be tested as specified in MIL-STD-200, method 110 except that steps 1 through 4 shall be deleted from the procedure and parts I and II below shall apply instead. No electrical load shall be connected to the terminals during this test.

Part I. The specimen shall be placed on a test chamber which has been vented to the atmosphere. The internal temperature of the chamber shall be maintained at 25°C (77°F) with a relative humidity that shall not exceed 30 percent at any time during test. The sand and dust density shall be raised and maintained at 0.1 to 0.5 gram per cubic foot within the test space. The sand and dust velocity through the test chamber shall be 2500 ± 500 feet per minute. This portion of the test shall be continued for 6 hours.

Part II. At the completion of the 6-hour period stated in part I the temperature shall then be raised to and maintained at 71°C (160°F) for an additional 6 hours with all other conditions remaining the same as in part I.

4.8.3 Fungus resistance. To determine conformance to 3.4.3, the switch shall be subjected to the fungus resistance test of MIL-F-13927. Fungal incubation shall be for 90 days per method B for class 2 specimens.

4.8.4 Waterproofness. To determine conformance to 3.4.4, the switch shall be tested by submersion as specified in MIL-STD-1184 for type II, class 2 components. Verify that the switch is operable and leakproof.

4.8.5 Temperature resistance. To determine conformance to 3.4.5, the swtich shall be tested at extremes of low and high temperature as specified in 4.8.5.1 and 4.8.5.2. During each of these tests the switch shall be energized by a 28 V dc source and connected to the rated loads specified on the applicable drawing. The switch shall be operated through its entire range 1000 times. For lamp loads, the load cycle shall be 1 ± 0.1 seconds "on" and 5 ± 0.1 seconds "off" (10 cycles per minute). The inductive load shall be 3 ± 0.1 seconds "on" and 3 ± 0.1 seconds "off" (10 cycles per minute). The switch shall then be tested for operating, torque per 4.7.5(a). These tests shall be performed in a chamber capable of maintaining air temperatures of minus 65°F and plus 165°F. Air movement during high temperature exposure shall be held to a minimum to prevent a temperature gradient and any cooling resulting from excess air motion.

4.8.5.1 Low temperature. To determine conformance to 3.4.5.1, the switch shall be placed in a cold chamber and conditioned for 24 hours prior to testing at minus 65 ± 5°F. Then, with the chamber temperature constant at minus 65 ± 5°F, the switch shall be operated and tested as specified in 4.8.5. Verify that operating torque conforms to 3.4.5.1 and that there is no degradation of electrical and mechanical performance.

4.8.5.2 High temperature. To determine conformance to 3.4.5.2, the switch shall be connected, conditioned and tested as specified in 4.8.5.1, except that constant temperature shall be plus 165 ± 5°F.

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5. PACKAGING

5.1 Preservation, packaging, packing, and marking. Preservation, packaging, packing, and marking for the desired level shall be in accordance with the applicable packaging standard or packaging data sheet specified by the contracting authority (see 6.2).

6. NOTES

6.1 Intended use. Rotary switches covered by this specification are intended primarily for use on tactical vehicles.

6.1.1 Class 3 switches. Switches that are non-waterproof, except that operating shaft resists entry of water, are for installation as components of sealed, waterproof assemblies containing non-waterproof items enclosed in waterproof housings.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Type and class of switch (see 1.2).
c. If first article inspection is not required or is other than as specified (see 3.1 and 4.4).
d. Applicable drawings if required (see 3.2).
e. If color other than specified is required (see 3.5.1).
f. Any special plating requirements (see 3.5.4).
g. If inspection conditions are other than as specified (see 4.3).
h. Selection of applicable level and packaging standard or packaging data sheet (see 5.1).

6.3 Definition.

6.3.1 Voltage drop. When the switch is built with permanently attached flexible leads (pigtails) and the switch terminals have been sealed (as with a molded sealant or potting compound), it may be necessary to measure total conductor terminal or conductors terminal drop and make necessary allowance for conductor voltage drop (see 3.3.1 and 4.7.1).

6.4 Switch lever. A lever is not supplied as part of the switch. The lever required will be determined by type of engine (gasoline or multifuel) on which the switch is used (see 3.2).

6.5 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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6.6 Subject term (key word) listing.

Switches, rotary
Rotary switches
28 V dc rotary switches
Switches, single-pole, single throw
Switches, single-pole, multi-throw
Switches, multipole, single throw
Switches, multipole, multi-throw
Switches, waterproof
Switches, non-waterproof
Switches, non-waterproof, with sealed (waterproof) shaft

6.7 Recycled materials. The use of recycled materials which meet the requirements of the applicable material specifications without jeopardizing the intended use shall be encouraged (see 3.2).

Custodian:
Army - AT

Review activity:
Air Force - 85

User activity:
Army - ME

Preparing activity:
Army - AT

(Project 5930-1337)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-S-13623 C	SWITCHES, ROTARY, 28 VOLT D.C.

3a. NAME OF SUBMITTING ORGANIZATION	4 TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*

5 PROBLEM AREAS

a. Paragraph Number and Wording

b Recommended Wording.

c Reason/Rationale for Recommendation

6 REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8 DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426

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